Chapter 7 Stats Lab

Instructions

Work through the following prompts and respond to each question in the space following each question.
To submit your work render the file by clicking the blue segmented arrow at the top of this window to render to a .html. Submit the resulting "Chapter 7 Stats Lab.html" and "Chapter 7 Stats Lab_files" folder to the Google drive folder I shared.

Background

In this Lab, we will work on applying the Central Limit Theorem for Means and Sums to a provided dataset. For this lab, I chose "The City of Pittsburgh Trees", a dataset that includes information on the over 45000 trees managed by the City of Pittsburgh Department of Public Works Forestry Division.

Getting Started

1. Run the following code block to load the dataset:

```
# Check if used packages are installed, install if not
if(!require(tidyverse, quietly = TRUE)) {
  install.packages("tidyverse")
  require(tidyverse)
}
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.1     v tibble    3.2.1
v lubridate 1.9.4     v tidyr     1.3.1
```

```
v purrr     1.0.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becon
```

```r
# Check if data has already been downloaded, download if not.
if(!file.exists("trees.csv")) {
  download.file("https://data.wprdc.org/datastore/dump/1515a93c-73e3-4425-9b35-1cd11b2196da"
}

# Check if data has been loaded to RStudio, load if not
if(!exists("Pittsburgh_trees")) {
  # Load data file
  trees <- read.csv("trees.csv")
  # Select height data (ignore NA values)
  tree_height <- data.frame(
    height = trees$height[!is.na(trees$height)])
  # Select estimated $ benefit data (ignore NA values)
  tree_benefits <- data.frame(
    benefits = trees$overall_benefits_dollar_value[!is.na(trees$overall_benefits_dollar_valu
}
```

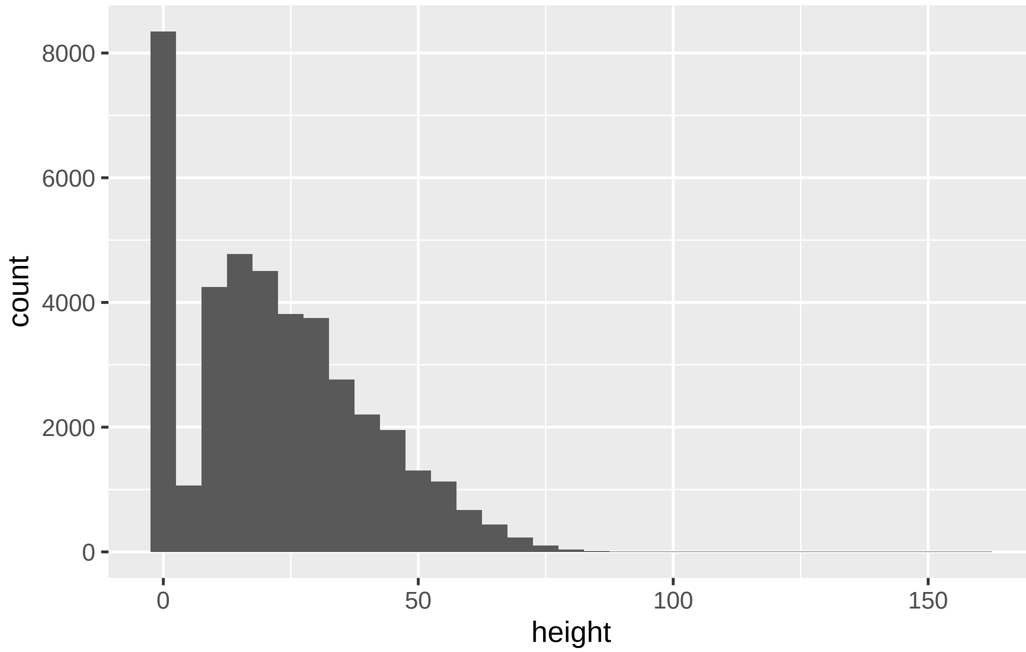
2. Follow the guided exercises in this notebook, running code chunks as you encounter them.
3. Respond to all Questions inside callouts.

Explore the data

Tree Heights

Let's take a look at the distribution of data by constructing a histogram:

```r
if(!require(ggplot2, quietly = TRUE)) {
  install.packages("ggplot2")
  require(ggplot2)
}

ggplot(data = tree_height, aes(x = height)) +
  geom_histogram(binwidth = 5)
```



There seems to be something strange going on, there are a lot of trees with a low height! Before going further, let's try to figure out what is going on.

Why low heights?

From the graph, the strange spike in low heights seems to be less than 10, so let's look at a summary of just those values.

```
tree_height %>%
  filter(height < 10) %>% # Look only at heights less than 10
  group_by(height) %>% # Group by distinct heights
  summarize(n = n()) # Count number of each distinct height
```

```
# A tibble: 10 x 2
   height     n
    <int> <int>
 1      0  8328
 2      1     2
 3      2    13
 4      3    52
 5      4    70
 6      5   175
 7      6   314
```

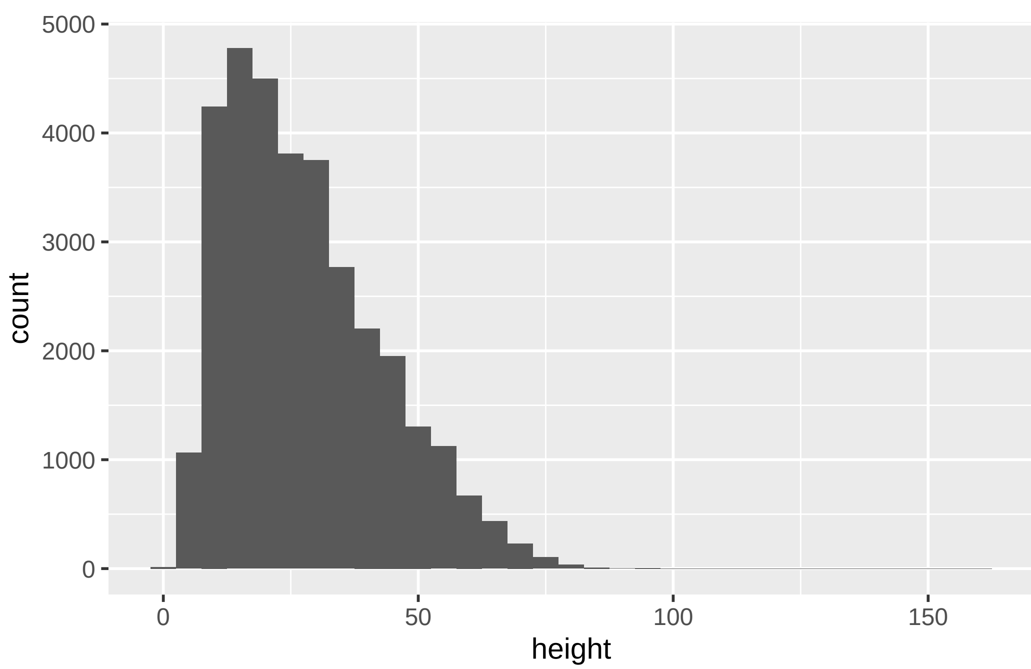
```
 8         7    452
 9         8    725
10         9    505
```

> **Question**
>
> How many trees have a height of 0?

Depending on the exact question you are trying to answer we may want to include trees with height 0 or exclude them. For this lab, lets narrow the data to only include trees with a recorded height. Note that if this were a report or scientific paper, you would have to fully disclose your selection criteria and reasoning behind it.

In this set, 1043 trees with height 0 are stumps, but also 2247 trees are listed as having height 0 but condition "Good". Given the unclear nature of trees of height 0, a reasonable choice for our class exercise is to rephrase questions about tree height to be about "trees with a recorded height".

```r
# Remove trees with height == 0
tree_height <- tree_height %>% filter(height != 0)

# Check histogram
ggplot(data = tree_height, aes(x = height)) +
  geom_histogram(binwidth = 5)
```

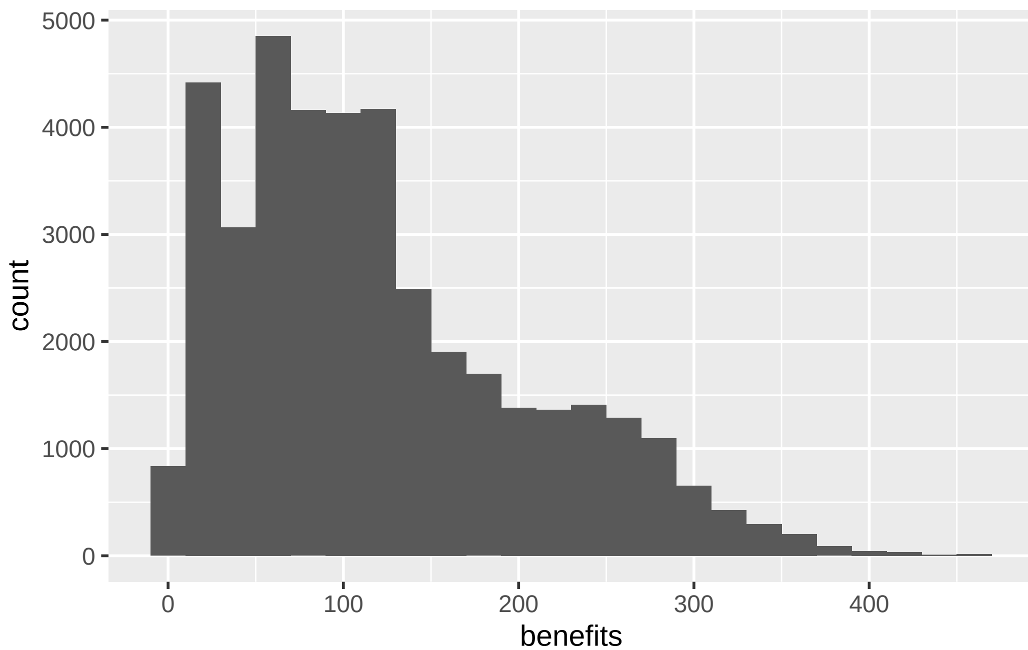


> **i** Question
>
> Does the height of trees with recorded heights in this data set appear to be normally distributed?

Tree Economic Benefit value

Let's also look at the distribution of estimated tree benefit value by constructing another histogram:

```
ggplot(data = tree_benefits, aes(x = benefits)) +
  geom_histogram(binwidth = 20)
```



> **i** Question
>
> Are there any strange values in the `tree_benefits` data?
> Do tree economic benefits in this data set appear to be normally distributed?

Central Limit Theorem (CLT)

Recall that the CLT says that the number of samples depends on whether the *original population* is normally distributed. Since neither of these are normally distributed, we should pick sample sizes of $n = 30$ or larger.

> **Tip**
>
> Central Limit Theorem of Means:
>
> $$\bar{X} \sim N\left(\mu_x, \frac{\sigma_x}{\sqrt{n}}\right)$$

> **Tip**
>
> Central Limit Theorem of Sums:
>
> $$\Sigma X \sim N\left(n \cdot \mu_x, \sqrt{n} \cdot \sigma_x\right)$$

CLT for Means - Tree Height

Next let's explore how the CLT for means works for our data. To do this, we'll sample the data $n = 30, 60$, and 150 times. For each of the different sample sizes, the sample will be taken (with replacement), the mean found, and the procedure repeated 10,000 times. This will give us a good picture of the CLT for the means.

```
# Repeat (replicate) finding the mean of 30 samples a total of 10,000 times
times <- 10000
CLT_Mean_Height_30 <- replicate(n = times,
          mean(sample(x = tree_height$height,
                      size = 30,
                      replace = TRUE))
)

# Find the mean and sd for n=30
mean30 <- mean(CLT_Mean_Height_30)
sd30 <- sd(CLT_Mean_Height_30)

# Repeat (replicate) finding the mean of 60 samples a total of 10,000 times
CLT_Mean_Height_60 <- replicate(n = times,
          mean(sample(x = tree_height$height,
```

n	mean	sd
30	27.74033	2.782582
60	27.73495	1.925600
150	27.75643	1.218108

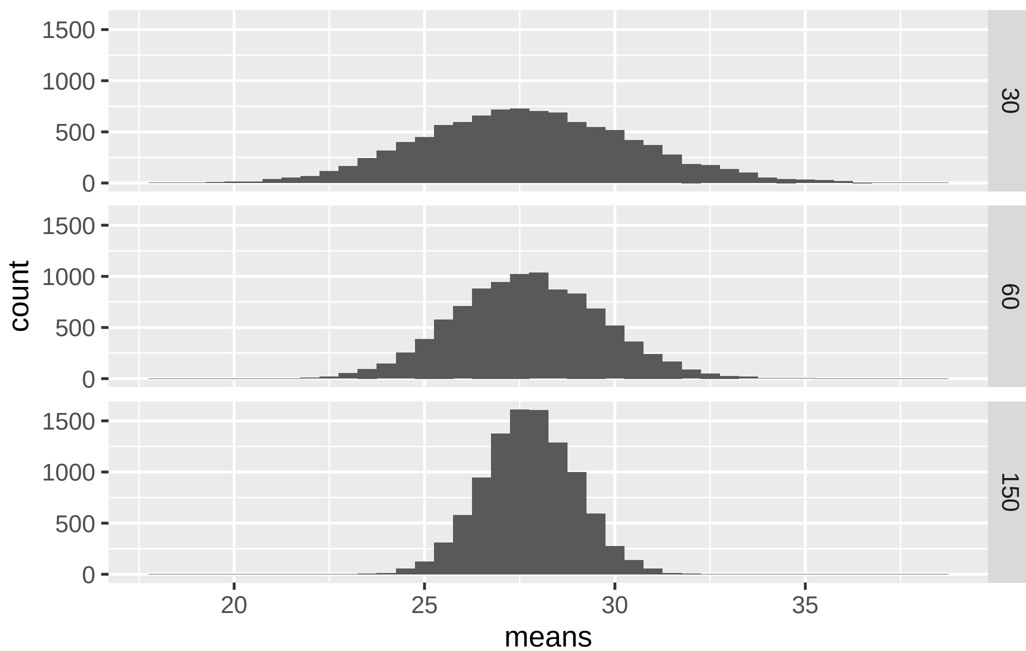
```
                      size = 60,
                      replace = TRUE))
)

# Find the mean and sd for n=60
mean60 <- mean(CLT_Mean_Height_60)
sd60 <- sd(CLT_Mean_Height_60)

# Repeat (replicate) finding the mean of 150 samples a total of 10,000 times
CLT_Mean_Height_150 <- replicate(n = times,
          mean(sample(x = tree_height$height,
                      size = 150,
                      replace = TRUE))
)

# Find the mean and sd for n=150
mean150 <- mean(CLT_Mean_Height_150)
sd150 <- sd(CLT_Mean_Height_150)

if(!require(gt, quietly = TRUE)) {
  install.packages("gt")
  require(gt)
}

data.frame(n = c(30,60,150),
           mean = c(mean30, mean60, mean150),
           sd = c(sd30, sd60,sd150)
) %>%
  gt::gt(caption = "Mean and Standard Deviation of Mean for different sample sizes n")
```

Question

What do you notice about the mean and standard deviation as the number of samples increases?

Let's look at a histogram of the means as the number of samples increases:

```
# Create data frame to graph the sampled means
CLT_M <- data.frame(
  samples = rep(c(30,60,150), each = times),
  means = c(CLT_Mean_Height_30,
                CLT_Mean_Height_60,
                CLT_Mean_Height_150))

# Create histograms of sample means
CLT_M %>%
  ggplot(aes(x = means)) +
  geom_histogram(binwidth = .5) +
  facet_grid(vars(samples))
```



> **Question**
>
> As n changed, why did the shape of the distribution change? Use 1-2 sentences to describe what happened.
> Answer:

Question

Suppose you were asked the following: "If you sample the height of 60 trees managed by the City of Pittsburgh (for which heights are recorded), what is the probability that the mean height will be less than 25 feet?"
Use the following function to answer the question:

```
p <- 0 # Change me!
mean <- 0 # Change me!
sd <- 1 # Change me!
pnorm(p, mean, sd)
```

```
[1] 0.5
```

Answer:

CLT for Sums - Tree Economic Benefits

Finally let's explore how the CLT for sums works for our data. To do this, we'll again sample the data $n = 30, 60,$ and 150 times. For each of the different sample sizes, the sample will be taken (with replacement), the *sum* found, and the procedure repeated 10,000 times. This will give us a good picture of the CLT for the sums.

```
# Repeat (replicate) finding the sum of 30 samples a total of 10,000 times
times <- 10000
CLT_Sum_Ben_30 <- replicate(n = times,
          sum(sample(x = tree_benefits$benefits,
                      size = 30,
                      replace = TRUE))
)

# Find the mean and sd for n=30
mean30 <- mean(CLT_Sum_Ben_30)
sd30 <- sd(CLT_Sum_Ben_30)

# Repeat (replicate) finding the mean of 60 samples a total of 10,000 times
CLT_Sum_Ben_60 <- replicate(n = times,
          sum(sample(x = tree_benefits$benefits,
                      size = 60,
                      replace = TRUE))
)
```

n	sum_means	sum_sd
30	3631.669	467.2503
60	7289.783	665.6399
150	18220.669	1038.1276

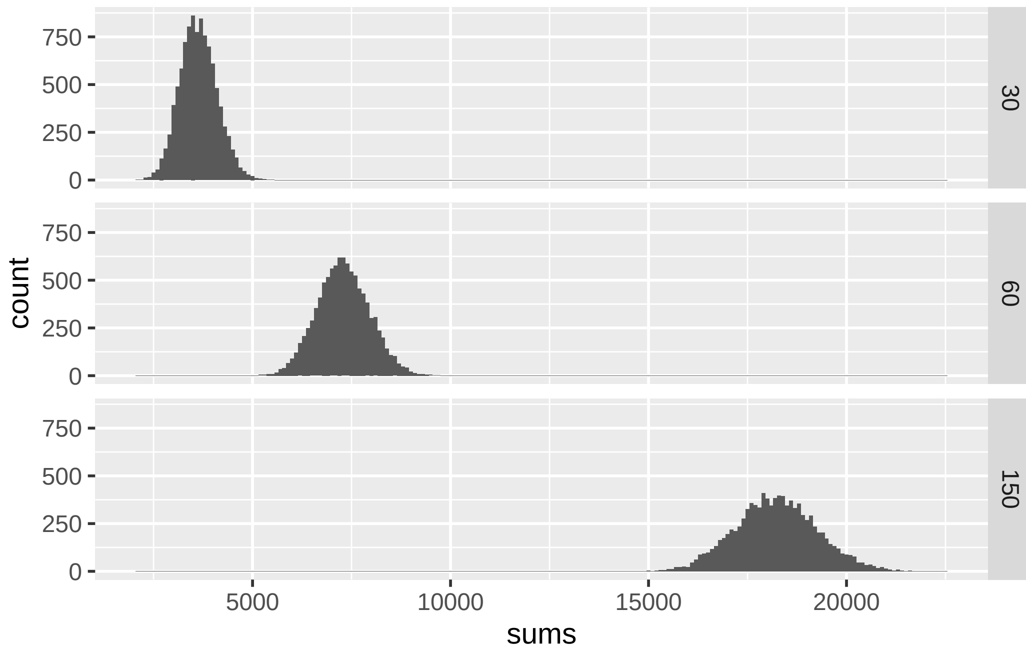
```
# Find the mean and sd for n=60
mean60 <- mean(CLT_Sum_Ben_60)
sd60 <- sd(CLT_Sum_Ben_60)

# Repeat (replicate) finding the mean of 150 samples a total of 10,000 times
CLT_Sum_Ben_150 <- replicate(n = times,
          sum(sample(x = tree_benefits$benefits,
                      size = 150,
                      replace = TRUE))
)

# Find the mean and sd for n=150
mean150 <- mean(CLT_Sum_Ben_150)
sd150 <- sd(CLT_Sum_Ben_150)

data.frame(n = c(30,60,150),
           sum_means = c(mean30, mean60, mean150),
           sum_sd = c(sd30, sd60,sd150)
) %>%
  gt::gt(caption = "Mean and Standard Deviation of Sum for different sample sizes n")
```

> **Question**
>
> What do you notice about the mean and standard deviation as the number of samples increases?

Let's look at a histogram of the means as the number of samples increases:

```
# Create data frame to graph the sampled means
CLT_S <- data.frame(
  samples = rep(c(30,60,150), each = times),
  sums = c(CLT_Sum_Ben_30,
                CLT_Sum_Ben_60,
                CLT_Sum_Ben_150))
```

```r
# Create histograms of sample means
CLT_S %>%
  ggplot(aes(x = sums)) +
  geom_histogram(binwidth = 100) +
  facet_grid(vars(samples))
```



> **Question**
>
> As n changed, why did the shape of the distribution change? Use 1-2 sentences to describe what happened.
> Answer:

> **Question**
>
> Suppose you were asked the following: "If you add the estimated benefit value of a random group of 150 trees in Pittsburgh, what is the probability that their overall value is greater than $17,000?"
> Use the following function to answer the question:

```r
p <- 0 # Change me!
mean <- 0 # Change me!
sd <- 1 # Change me!
lower <- FALSE
pnorm(p, mean, sd, lower.tail = lower)
```

```
[1] 0.5
```

Answer: